

Assignment #2

**Central limit theorem. Probability distribution: Binomial,
Poisson, Uniform, Normal, Exponential**

1.	(i) If $n = 5$, $p = 20$ then what will be the mean and variance of Binomial distribution. (ii) The mean and variance of Binomial distribution are 4 and $4/3$ respectively. Find $P(X \geq 1)$. (Ans. 0.99863) (iii) Comment on the following statement: "The mean of a Binomial distribution is 5 and the standard deviation is 3". (iv) If X is a Poisson variate such that $P(X = 2) = 9P(X = 4) + 90P(X = 6).$ find the mean and variance of X. (Ans.: Mean = variance = 1) (v) If $n = 4$ and $p = 25$ then the mean and variance of Poisson distribution will be..... (vi) Under normal curve total area right side of mean is... (Ans. 50%) (vii) Mean and variance of standard normal variate are..... (viii) What is the p.d.f. for Uniform distribution. (ix) The mean and variance of Uniform distribution are 1 and $4/3$ respectively. Find $P(X < 0)$. (x) What is the p.d.f. of exponential distribution. (xi) In which condition mean and variance of exponential distribution are equal. (xii) Which continuous distribution follows memoryless property.
2.	Out of 800 families with 4 children each, how many families would be expected to have (i) 2 boys and 2 girls (ii) at least one boy (iii) no girl (iv) at most two girls? (Ans. 300, 750, 50, 550)
3.	Twenty percent screw produced in a certain factory turn out to be defective. Find the probability that in a sample of 15 screws chosen at random exactly four will be defective.
4.	The probability that a bomb dropped from a plane will strike the target is $1/5$. If six bombs are dropped, find the probability that at least two will strike the target and exactly two will strike the target. (Ans. 0.34464, 0.24576)
5.	In a book of 520 pages, 390 typo-graphical errors occur. Assuming Poisson law for the number of errors per page, find the probability that a random sample of 5 pages will contain no error. (Ans. $e^{-3.75}$)
6.	Six coins are tossed 6400 times. Using the Poisson distribution, find the approximate probability of getting 6 heads 2 times. (Ans. $e^{-100}5000$)
7.	A box contains 100 cellphones, 20 of which are defective. Ten cellphones have been selected for inspection. Find the probability that (i) at least one is defective (ii) at the most three are defective (iii) all are defective (iv) none is defective.
8.	In a certain examination 2000 students appeared. The average marks obtained were 50% and standard deviation was 5%. How Many students do you expect to obtain more than 60 marks, assuming the marks to be distributed normally.? (Ans. 46 marks)

9.	Subway trains on a certain line run every half hour between midnight and six in the morning. What is the probability that a man entering the station at random time during this period will have to wait at least twenty minutes? (Ans. $\frac{1}{3}$)
10.	A random sample of size 100 is taken from an infinite population having mean 76 and variance 256. What is the probability that sample mean will be between 75 and 78? (Ans. 0.6311)
11.	A car hire firm has two cars, which it hires out day by day. The number of demands for a car on each day distributed as a Poisson distribution with mean 1.5, calculate the proportion of days on which (i) neither car is used, and (ii) the proportion of days on which some demand refuses.
12.	Assume mean height of soldiers to be 68.22 inches with a variance of 10.8 inches square. How many soldiers in a regiment of 1000 would you expect to be over 6 feet tall, given that the area under the standard normal curve between $x = 0$ and $x = 0.35$ is 0.1368 and between $x = 0$ and $x = 1.15$ is 0.3746. (Ans. 125(approx.))
13.	If X is Uniformly distributed in $[-2, 2]$, find (i) $P(X < 0)$ (ii) $P[X-1 \geq \frac{1}{2}]$ (Ans. $\frac{1}{2}, \frac{3}{4}$)
14.	The time (in hours) required to repair a machine is exponentially distributed with parameter $\lambda = \frac{1}{3}$. What is the probability that the repair time exceeds 3 hours? (Ans. 0.3679)
15.	The life length (in months) X of an electric component follows an exponentially distribution with parameter $\lambda = \frac{1}{2}$. What is the probability that the component survives at least 10 months given that already it had survived for more than 9 months? (Ans. 0.6065)
16.	A random sample of size 100 is taken from a population whose mean is 60 and variance is 400. Using CLT, with what probability can we assert that the mean of the sample will not differ from $\mu = 60$ by more than 4?
17.	In a normal distribution, 31% of items are under 45 and 8% are over 64. What is the mean and standard deviation of the distribution? (Given that $\phi(1.41) = 0.42$, $\phi(0.19) = 0.50$) (Ans. mean = 50, S.D. = 10)
18.	A coin is tossed 12 times. Find the probability, both exactly and by fitting a normal distribution of getting (a) 4 heads (b) at most 4 heads. (Given that $\phi(1.44) = 0.4251$, $\phi(0.866) = 0.051$) (Ans (a) 0.12, (b) 0.1949)
19.	The number of personal computers (PCs) sold daily at Computer World is Uniformly distributed with a minimum of 2000 PCs and a maximum of 5000 PCs. (i) Find the probability that the daily sales will fall between 2500 and 3000 PCs. (ii) What is the probability that Compu World will sell at least 4000 PCs and exactly 2500 PCs? (Ans. 0.1667, 0.3333, 0)
20.	The length of a time a person speaks over phone follows exponential distributions with mean 6. What is the probability that the person will talk for (i) more than 8 minutes (ii) between 4 and 8 minutes? (Ans. 0.2636, 0.2498)